Nick Kunz

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Research Interests

My research interests are at the intersection of data science, machine learning, and complex systems. Many of the questions I ask are motivated by understanding emergent dynamics under evolving real-world constraints. My work has direct applications in urban planning, economics, defense, and other areas of societal impact.

Education

Cornell University — Ithaca, NY 2027
Ph.D., Systems Engineering, Artificial Intelligence
Advisors: H. Oliver Gao, Francesca Parise, Wenfei Xu

Cornell University — Ithaca, NY 2025
M.S., Systems Engineering
Advisor: H. Oliver Gao

Columbia University — New York, NY 2019
M.S., Urban Planning, Urban Analytics
Advisors: Lance Freeman, Kazuki Sakamoto

Harvard University — Cambridge, MA 2012
Non-Degree, Urban Planning
Advisor: Nick Hornig

Arizona State University — Tempe, AZ 2012
B.S. (Hons.), *summa cum laude*, Housing and Urban Development
Advisors: Kevin Kellogg, Michael Underhill, Trevor Barger

Publications

Journal Articles

N. Kunz, H. O. Gao, "A General Capacity Frontier of Complex Networks". *Preprint arXiv:2610.09574*. 2026. [Link]

N. Kunz, H. O. Gao, "Global Geolocated Realtime Data of Urban Transit Bus Idling". *Preprint arXiv:2403.03489*. 2026. [Link]

D. Lu, L. Cui, **N. Kunz**, *et al.*, "Urban Infrastructure and Spatiotemporal Environmental Features for EGFR-Mutant Lung Cancer". *Preprint medRxiv:18.26353481v1*. 2026. [Link]

C. Avci, C. Huang, **N. Kunz**, H. O. Gao, "uTECH-Population: Synthetic Population Generation with Housing Assignment". *Under Review*. 2026.

Conference Proceedings

W. Wu, **N. Kunz**, P. Branco, "ImbalancedLearningRegression" in *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD, September 19–23, 2022, Grenoble, France,* Proceedings, Part VI, Lecture Notes in Computer Science, vol 13718. Springer, pp. 645–648, 2023. [Link]

Working Papers

N. Kunz, H. O. Gao, "Global Geolocated Historical Data of Urban Transit Bus Idling From 2024 to 2026". *In Preparation*. 2026.

N. Kunz, H. O. Gao, "Network Constrained Spatial Prediction: Transfer Learning Geolocations". *In Preparation*. 2026.

Theses

N. Kunz, "Structural Systems Intelligence: Network-Constrained Estimation of Global Transit Dynamics," Ph.D. Dissertation, Duffield College of Engineering, Cornell Univ., Ithaca, NY, 2027.

N. Kunz, "Unsupervised Learning for Submarket Modeling: A Proxy for Neighborhood Change," M.S. Thesis, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available: Columbia Academic Commons. [Link]

N. Kunz, "Realizing Interactive Architecture: A Driver of the Knowledge Economy," B.S. (Hons.) Thesis, Barrett, The Hons. Col., Arizona State Univ., Tempe, AZ, 2012. Available: Barrett, The Honors College Thesis/Creative Project Collection. [Link]

Industry Reports

N. Kunz, "n-Dimensional Mapping by Normalized Value Approximation: Squad Immersive Virtual Trainer (SiVT) Safety Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, "Normalized n-Dimensional Inverse Euclidean Distance: Combining Independent Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, "Minimum-Weighted Mean Variability: Within Group Performance Metrics," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, "An Exploratory Data Analysis of the Gavin Cup: Bringing Readiness & Performance Metrics Into Focus," Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

Software Packages

GRD-TRT-BUF-4I: Ground Truth Buffer for Idling 2023 - 2027
GitHub [Link] arXiv [Link] Website [Link]

A novel algorithm designed as a series of scalable microservices that detects urban transit bus idling in realtime on global scale. Utilizes a combination of templated ETL pipelines to provide a comprehensive view of the emergent behavior of interfleet idling patterns across multiple transit networks around the world.

SMOGN: Synthetic Minority Over-Sampling Technique for Regression with Gaussian Noise 2021
GitHub [Link] PyPI [Link] Kaggle [Link] Stack Exchange [Link]

A pre-processing algorithm designed to address imbalanced data for regression problems. Conducts over-sampling with traditional interpolation, as well as with the introduction of Gaussian noise. Selects between the two over-sampling techniques by the KNN distances underlying a given observation.

NestedHyperBoost: Nested Cross-Validation for Bayesian Optimized Gradient Boosting 2020
GitHub [Link] PyPI [Link]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Gradient Boosting. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

NestedHyperLine: Nested Cross-Validation for Bayesian Optimized Linear Regularization 2020
GitHub [Link] PyPI [Link]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Linear Regularization. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

CRASSMAT: Conditional Random Sampling Sparse Matrices 2019
GitHub [Link] CRAN [Link]

A novel matrix sampling algorithm designed to conditionally random sample sparse matrices. Useful for training and test set splitting prior to model fitting in cross-validation procedures and for estimating the predictive accuracy of imputation methods, such as NNMF and SVD.

Selected Projects

N. Kunz, S. Huang, S. Osakata, "Cloudy Night: Activating Park Space by Illuminated Interactivity," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

GSAPP [Link] GitHub [Link] YouTube [Link] Faculty [Link]

N. Kunz and S. Bhele, "Signal or Noise? Qualified Opportunity Zone Effect on Development Activity in Queens," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

GSAPP [Link] GitHub [Link] Issuu [Link]

E. Hudgins and **N. Kunz**, "Planning for Extreme Settlements, McMurdo Station, Antarctica," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

GSAPP [Link] Issuu [Link]

J. Karasik, S. Lee, **N. Kunz**, *et al.*, "Negotiating Fragments (Restructuring Quarto Workshop)," Genoa, Italy, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

GSAPP [Link] YouTube [Link] Issuu [Link] Issuu [Link] Press [Link]

N. Kunz, "Proportionalities: A Decade of Housing Markets to Subway Stations in Manhattan from 2006 to 2016," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

GSAPP [Link] GSAPP [Link] Issuu [Link]

N. Kunz, "Reimagined: John Snow's Cholera Map of 1854," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

Issuu [Link]

G. S. Coch, S. Nelsen, **N. Kunz**, K. Warner, "Seeking Refuge: The Geography of Migrant Deaths Along the US-MX Border," Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

GSAPP [Link] Medium [Link]

N. Kunz, "Health & Wellness Alliance for Children: Children's Asthma, Healthy Physical Environments," Dallas County, Texas, bcWORKSHOP, Dallas, TX, 2014. [Online]. Available:

Issuu [Link]

N. Kunz, "The Bexar Street Corridor: A Century in Development," Dallas, Texas, bcWORKSHOP, Dallas, TX, 2013. [Online]. Available:

Issuu [Link] Issuu [Link] bc [Link] Vimeo [Link]

N. Kunz, "Fenway-Kenmore Square: Comprehensive Planning & Finance," Cambridge, Massachusetts, Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

Issuu [Link]

N. Kunz, "Design Representation: Analog Modeling & Hand Drafting," Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

Issuu [Link]

N. Kunz, "Danelle Plaza: Complete Street Mobility Planning," Tempe, Arizona, Herberger Inst. for Des. and the Art., Arizona State Univ., Tempe, AZ, 2012. [Online]. Available:

Issuu [Link]

N. Kunz, "Ascension: Multifamily Mixed-Use Real Estate Development Proposal," Tempe, Arizona, Independent Project, Phoenix, AZ, 2012. [Online]. Available:

Issuu [Link]

N. Kunz, "Sui Residence: Second Story Architectural Design," Paradise Valley, Arizona, Independent Project, Phoenix, AZ, 2011. [Online]. Available:

Issuu [Link]

Professional Experience

Deloitte — Seattle, WA 2021 - 2025
Senior Data Scientist
Artificial Intelligence [Link]

Microsoft — Redmond, WA 2020 - 2021
Data Scientist
Integrated Visual Augmentation System [Link]

Pacific Prospecting Group — Seattle, WA 2017 - 2019
Data Scientist
Cryptocurrency Mining & Management [Link]

Brawner & Company — Snoqualmie, WA 2016 - 2017
Development Analyst
Real Estate Development [Link]

bcWORKSHOP — Dallas, TX 2013 - 2014
Fellow
Architecture & Planning [Link]

Mentorship Activities

Audrey Shin, B.S., Computer Science, Columbia University — New York, NY 2025 - 2026
Machine Learning Engineer Intern, ORO Labs
Professional development and management in early career engineering.

Albert Rosario-Pichardo, M.S., Urban Planning, Columbia University — New York, NY 2023 - 2024
GIS Analyst Intern, New York City Economic Development Corporation
Academic and professional development in early career public service.

Albert Ruiz Martinez, B.S., Computer Science, Arizona State University — Tempe, AZ 2023 - 2024
Analyst, Deloitte Consulting
Professional development and management in early career consulting.

Julia VanPutte, B.S., Operations Research, Cornell University — Ithaca, NY 2022 - 2024
Business Intelligence Engineer Intern, Amazon
Academic and professional development from undergraduate school to industry.

Chawner Hurd, M.B.A., Business, Columbia University — New York, NY 2020 - 2023
Associate Intern, Visa
Military transition from US Army, 5th Special Forces Group to graduate school and industry.

Michael Bollinger, B.A., Computer Science, Columbia University — New York, NY 2019 - 2023
Software Engineer Intern, Amazon
Military transition from US Army, 75th Ranger Regiment to undergraduate school and industry.

Daniel Hineline, B.A., Political Science, Columbia University — New York, NY 2018 - 2023
Associate Intern, Guggenheim Partners
Military transition from US Army, 82nd Airborne Division to undergraduate school and industry.

Funding & Support

Fellowships & Scholarships

Graduate Engineering Fellowship ($42,773), Cornell University — Ithaca, NY 2023
Systems Engineering Fellowship ($31,033), Cornell University — Ithaca, NY 2022
Korean Ancestry Grant ($8,000), William Orr Dingwall Foundation — Washington, DC 2010 - 2012
Barrett, The Honors College Scholarship ($4,000), Arizona State University — Tempe, AZ 2010 - 2012
AIERF Undergraduate Scholarship ($1,000), Appraisal Institute — Chicago, IL 2010 - 2011
Study.net Foundation Scholarship ($500), Study.net — Fort Lauderdale, FL 2010
Herberger Institute for Design & the Arts Scholarship ($300), Arizona State University — Tempe, AZ 2010
Center for Global Education Services Scholarship ($750), Arizona State University — Tempe, AZ 2009

Education Benefits & National Service Awards

Chapter 33: Post-9/11 GI Bill, US Dept. of Veterans Affairs — Washington, DC 2022 - 2026
Chapter 31: Veteran Readiness and Employment, US Dept. of Veterans Affairs — Washington, DC 2017 - 2019
Eli Segal AmeriCorps Education Award, US AmeriCorps — Washington, DC 2014

Honors & Awards

Applause Award, Deloitte Consulting — Seattle, WA [Link] 2022
Distinguished Alumni Award, Arizona State University — Tempe, AZ [Link] 2016

Talks & Presentations

Applications in Data Science, Cornell Data Science Club, Cornell University — Ithaca, NY 2026
Ground Truth Buffer for Idling, 3-Minutes of Madness, Cornell University — Ithaca, NY [Link] 2024
Global Geolocated Realtime Data, Flash Talks, Cornell University — Ithaca, NY 2023
Emerging Technology Workshop, Countryside Angels Talks, Ag Ventures Alliance — Mason City, IA 2021
Planning Paths in Private Tech, Alumni Panel, Columbia GSAPP — New York, NY [Link] 2021
Cloudy Night, End of Year Show, Columbia GSAPP — New York, NY [Link] 2019
Restructuring Quarto Workshop, Municipality of Genoa — Genoa, IT [Link] 2018
The Bexar Street Corridor, US Congresswoman Eddie Bernice Johnson's Office — Dallas, TX 2014
The Bexar Street Corridor, TR Hoover Community Development Corporation — Dallas, TX 2014
The Bexar Street Corridor, bcSHOPFRONT Series, bcWORKSHOP — Dallas, TX [Link] 2013
Realizing Interactive Architecture, Honors Symposium, Arizona State University — Tempe, AZ 2012

Media & Press

Urban Planning Semester In Review: Alumni Updates, Columbia GSAPP — New York, NY [Link] 2023
Spotlights: Nick Kunz, Cornell Systems Engineering — Ithaca, NY [Link] 2023
Meet the N-K Alumni Nick Kunz, Northwood Anchor, Vol. 126, No. 41, Pg. 4 — Northwood, IA [Link] 2022
With Special Guest, Nick Kunz, TWA Family Hour Podcast — Seattle, WA [Link] 2020
Restructuring Quarto Workshop, Municipality of Genoa Press — Genoa, IT [Link] 2018
Achievement Oriented: Nick Kunz, ASU Magazine, Vol. 20, No. 2, Pg. 7 — Tempe, AZ 2016
Welcome Nick Kunz, bcWORKSHOP — Dallas, TX 2013

Academic Service

Tuesday Talks Mentorship, Columbia GSAPP — New York, NY [Link] 2026
Alumni Mentorship Program, Columbia GSAPP — New York, NY [Link] 2023 - 2024

Military Service

Assignments

US Army, 2nd Ranger Battalion, 75th Ranger Regiment — Fort Lewis, WA 2015 - 2016
Infantry

US Army, Ranger Training Detachment — Fort Benning, GA 2015
Ranger Assessment and Selection Program

US Army, 1st Special Warfare Training Group — Fort Bragg, NC 2014
Special Forces Assessment and Selection

US Army, 1st Special Warfare Training Group — Fort Bragg, NC 2014
Special Forces Preparation and Conditioning

US Army, 507th Parachute Infantry Regiment — Fort Benning, GA 2014
Airborne School

US Army, 19th Infantry Regiment — Fort Benning, GA 2014
Infantry School

Decorations

Expert Badge, Rifle Component Bar, US Army, 75th Ranger Regiment — Fort Lewis, WA 2016
Global War on Terrorism Service Medal, US Army, 75th Ranger Regiment — Fort Lewis, WA 2016
Driver Badge, Wheeled Vehicle Component Bar, US Army, 75th Ranger Regiment — Fort Lewis, WA 2015
Parachutist Badge, Unit Trimming, US Army, 507th Parachute Infantry Regiment — Fort Benning, GA 2014
Infantry Blue Cord Fourragère, US Army, 19th Infantry Regiment — Fort Benning, GA 2014
National Defense Service Medal, US Army, 19th Infantry Regiment — Fort Benning, GA 2014
Army Service Ribbon, US Army, 19th Infantry Regiment — Fort Benning, GA 2014

Civil Service

AmeriCorps — Dallas, TX 2013 - 2014
Volunteer in Service to America (VISTA)

Open Architecture Collaborative — Dallas, TX 2013 - 2014
Architecture & Planning

Trinity Park Conservancy — Dallas, TX 2013 - 2014
Public Park Conservation

Professional Memberships

Sigma Xi — Research Triangle Park, NC 2026 - Present
The New York Academy of Sciences — New York, NY 2024 - Present
AmeriCorps Alums of North Texas — Dallas, TX 2014 - Present
Barrett, The Honors College Alumni Association — Tempe, AZ 2012 - Present
US Military Veterans of Columbia University — New York, NY 2017 - 2018
Architecture for Humanity, Dallas/Fort Worth Chapter — Dallas, TX 2013 - 2014
Social Economic Environmental Design Network — Princeton, NJ 2013 - 2014
Urban Land Institute Arizona — Phoenix, AZ 2011 - 2012
United States Green Building Council, Central AZ Chapter — Phoenix, AZ 2009 - 2012
American Institute of Architecture Students — Washington, DC 2009 - 2010

Technical Skills

Languages

Scripting: Python, Bash
Statistical: R, Stata
Compiled: C/C++, Fortran
Database: SQL, NoSQL, Cypher
Typesetting: LaTeX, Markdown

Frameworks

Data Science: NumPy, SciPy, Pandas, statsmodels, Featuretools, Tidyverse
Visualization: Matplotlib, Seaborn, Plotly, Processing, Shiny, ggplot
Statistical Learning: Scikit-Learn, XGBoost, LightGBM, CatBoost
Reinforcement Learning: OpenAI Baselines, Stable Baselines
AI Engineering: Pinecone, LangChain, LangGraph
Deep Learning: PyTorch, TensorFlow, ONNX

Deployment

DevOps: Git, Ubuntu, Docker, CI/CD
Databases: PostgreSQL, SQLite, CosmosDB, Neo4j
Web Services: Flask, Gunicorn, Nginx
Web Automation: Selenium, Puppeteer
Cloud Platforms: Azure, AWS, Databricks
Security: STIGs, NIST, ZAP

Software

Development Environments: VS Code, RStudio, Jupyter
Geographic Information Systems: ArcGIS, QGIS, OSM, Leaflet, Mapbox
3D Modeling & Optimization: Rhino, Grasshopper, SketchUp, AutoCAD, Unity
Design & Illustration: Adobe Photoshop, Illustrator, InDesign
Financial Modeling: Excel, Macros, CoStar
Studio & Live Audio: Logic, Protools

Prototyping

Project Management: Agile, Scrum, Critical Path
Sensors & Hardware: Arduino, Raspberry Pi
Electrical: Soldering, Wiring, Safety
Analogs: Drawing, Hand Drafting, Physical Modeling

Personal Activities

Music & Live Audio

Atlas North American Tour — US & Canada 2013
The Word Alive *(Fearless Records)*, Co-Tour Manager
The Word Alive *(Fearless Records)*, Guitar Tech

Fight the Silence Tour — US 2013
The Word Alive *(Fearless Records)*, Guitar Tech
For Today *(Razor & Tie Records)*, Guitar Tech

All Stars Tour — US 2012
The Word Alive *(Fearless Records)*, Guitar Tech
A Skylit Drive *(Fearless Records)*, Guitar Tech

Vans Warped Tour — US & Canada 2011
The Word Alive *(Fearless Records)*, Co-Tour Manager
Of Mice & Men *(Rise Records)*, Guitar Tech

Regional Performances — US 2011
The Rendering *(Authentik Artists)*, Guitar

Updated: Oct. 2026